Comprehensive Perturb-seq Gene Set Analysis Report

Analysis Date: January 17, 2025
Output Directory: `results/perturbseq-analysis/`

Executive Summary

This report documents the analysis of **13 gene sets** across **2 Perturb-seq datasets** (RPE1 Essential, K562 GWPS). The analysis identifies genetic knockdowns that mimic or antagonize age-associated, AMD-associated, and pathobiological (Senescence, Redox, Inflammation) gene expression signatures.

Key Findings

1. **399 convergent genes** appear as top antagonists across multiple analyses
2. **Significant RPE1-K562 concordance** for Age (21 genes, p<1e-15) and Redox (11 genes, p<5.7e-6)
3. **High gene coverage** for Redox-PRO (94%) and Senescence-ANTI (92%)
4. **Lower coverage for AMD gene sets** (7-45%), reflecting tissue-specific expression

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Methods

Gene Sets Analyzed

Category	Gene Set	Size	Description
Age	Age-UP	323	Upregulated with age (GSE29801)
	Age-DOWN	161	Downregulated with age (GSE29801)
	ARE	242	NRF2/ARE targets (MSigDB + literature)
AMD	AMD-Macula-UP	126	Upregulated in AMD RPE Macula (p<0.01)
	AMD-Macula-DOWN	431	Downregulated in AMD RPE Macula (p<0.01)
	AMD-nonMacula-UP	395	Upregulated in AMD RPE non-Macula (p<0.01)
	AMD-nonMacula-DOWN	508	Downregulated in AMD RPE non-Macula (p<0.01)
Senescence	Senescence-PRO	250	Pro-senescence (SenMayo + GO + SASP)
	Senescence-ANTI	327	Anti-senescence (Cell cycle + E2F)
Redox	Redox-PRO	245	Pro-oxidative stress (HALLMARK + GO)
	Redox-ANTI	16	Antioxidant response (NRF2 targets)
Inflammation	Inflammation-PRO	571	Pro-inflammatory (HALLMARK + NFkB)
	Inflammation-ANTI	111	Anti-inflammatory (GO regulatory)

Perturb-seq Datasets

Dataset	Perturbations	Genes	Cell Type
RPE1 Essential	2,679	8,749	RPE1 (human RPE cells)
K562 GWPS	11,258	8,248	K562 (erythroleukemia)

Analytical Approach

Backward Analysis (Mimetic Scoring): For each knockdown, we computed a mimetic score:

$$\text{Mimetic Score} = \text{Enrichment(UP genes)} - \text{Enrichment(DOWN genes)}$$

- **Positive scores**: Knockdown upregulates UP genes and downregulates DOWN genes (mimetic)
- **Negative scores**: Knockdown has opposite effect (antagonist)

Results

Gene Set Coverage in Perturb-seq Data

Gene Set	RPE1 Coverage	K562 Coverage
Redox-PRO	94.7% (232/245)	93.9% (230/245)
Senescence-ANTI	91.7% (300/327)	92.7% (303/327)
ARE	62.4% (151/242)	52.9% (128/242)
Age-DOWN	49.7% (80/161)	46.6% (75/161)
Inflammation-PRO	46.9% (268/571)	35.9% (205/571)
Age-UP	43.7% (141/323)	36.5% (118/323)
AMD-Macula-UP	45.2% (57/126)	34.1% (43/126)
Senescence-PRO	43.2% (108/250)	31.6% (79/250)
Inflammation-ANTI	35.1% (39/111)	29.7% (33/111)
AMD-nonMacula-UP	32.7% (129/395)	24.6% (97/395)
AMD-nonMacula-DOWN	14.8% (75/508)	12.4% (63/508)
AMD-Macula-DOWN	7.4% (32/431)	7.7% (33/431)
Redox-ANTI	62.5% (10/16)	56.3% (9/16)

> [!NOTE] Low coverage for AMD-DOWN gene sets reflects tissue-specific expression in RPE that differs from cell line expression profiles.

Analysis Summary Statistics

Gene Set Pair	Dataset	UP Genes	DOWN Genes	Mean Score	Std	Range
Age	RPE1	141	80	-0.019	0.114	[-0.39, 0.31]
Age	K562	118	75	-0.027	0.150	[-0.46, 0.38]
AMD-Macula	RPE1	57	32	0.174	0.228	[-0.51, 0.60]
AMD-Macula	K562	43	33	0.021	0.220	[-0.55, 0.67]

Gene Set Pair	Dataset	UP Genes	DOWN Genes	Mean Score	Std	Range
AMD-nonMacula	RPE1	129	75	0.013	0.141	[-0.39, 0.36]
AMD-nonMacula	K562	97	63	-0.001	0.152	[-0.40, 0.40]
Senescence	RPE1	108	300	0.350	0.295	[-0.58, 0.83]
Senescence	K562	79	303	0.039	0.147	[-0.40, 0.67]
Redox	RPE1	232	10	0.007	0.281	[-0.65, 0.60]
Redox	K562	230	9	0.006	0.282	[-0.80, 0.68]
Inflammation	RPE1	268	39	0.014	0.122	[-0.34, 0.38]
Inflammation	K562	205	33	-0.005	0.170	[-0.43, 0.48]

RPE1 vs K562 Concordance

Gene Set	Overlap (top 200)	Jaccard	Hypergeom p-value	Spearman r	Status
Age	21	0.055	**1.0e-15**	0.153	YES Significant
Redox	11	0.028	**5.7e-6**	0.144	YES Significant
Inflammation	9	0.023	**1.9e-4**	-0.001	YES Overlap significant
AMD-Macula	6	0.015	0.015	0.064	Marginal
Senescence	5	0.013	0.051	0.264	Not significant
AMD-nonMacula	2	0.005	0.597	0.011	Not significant

> [!IMPORTANT] Age, Redox, and Inflammation show significant cross-dataset concordance, suggesting these pathways are conserved across cell types.

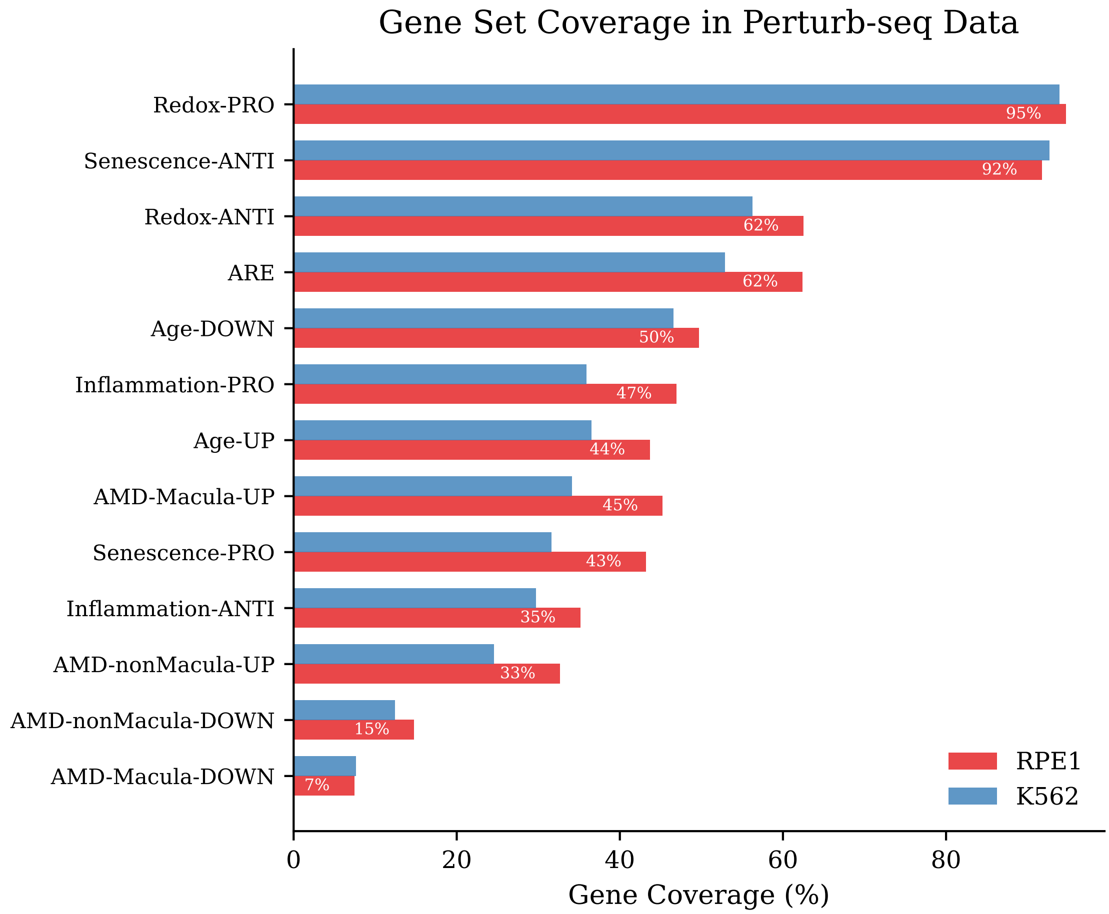


Figure 1: Results Coverage Summary

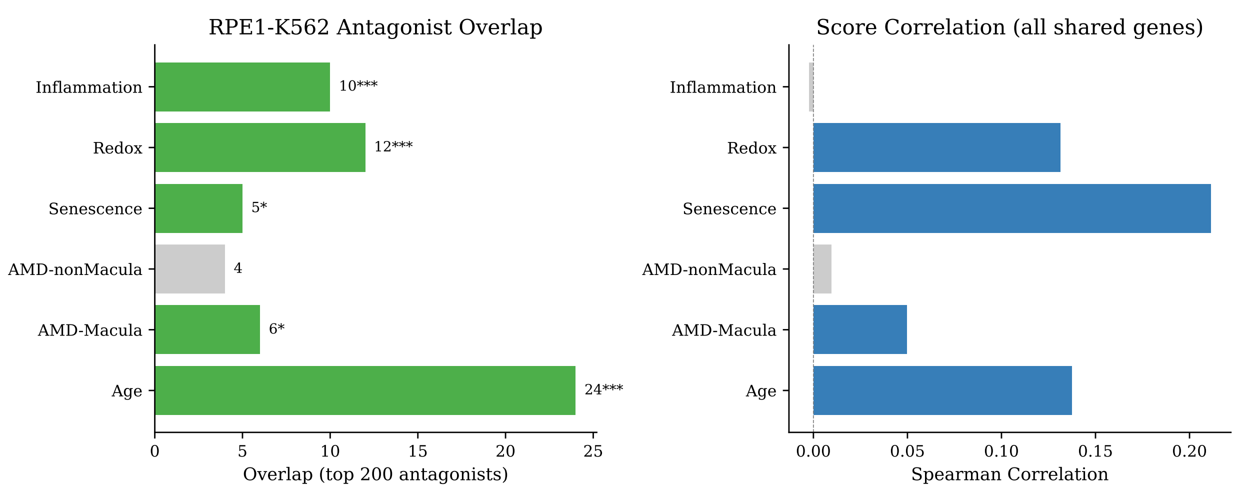
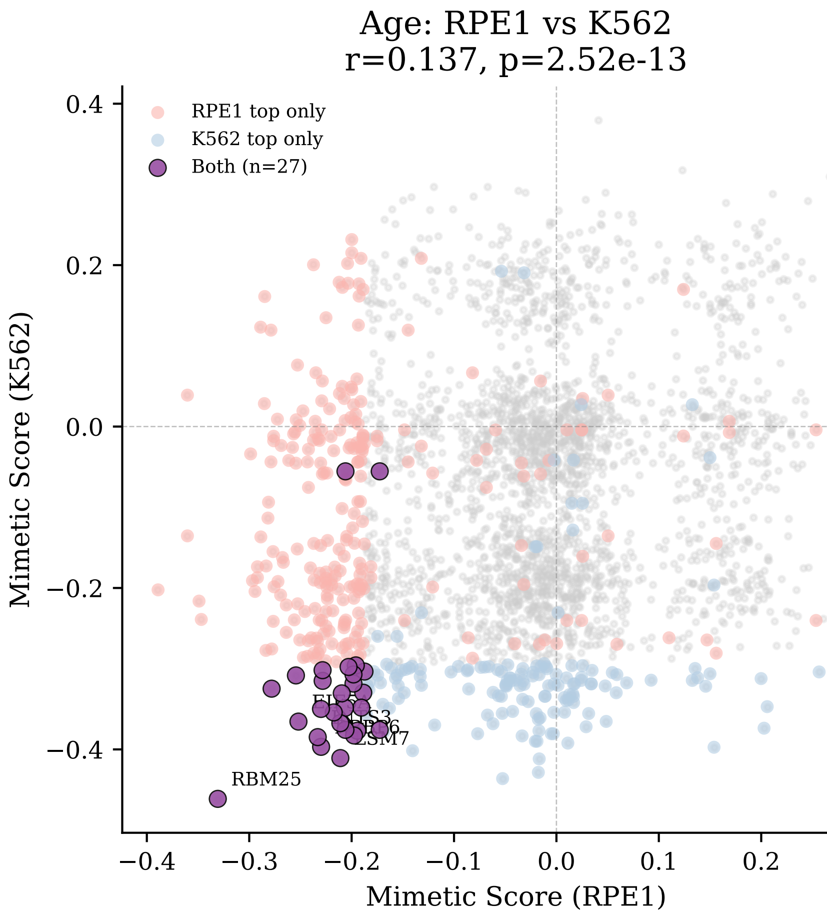
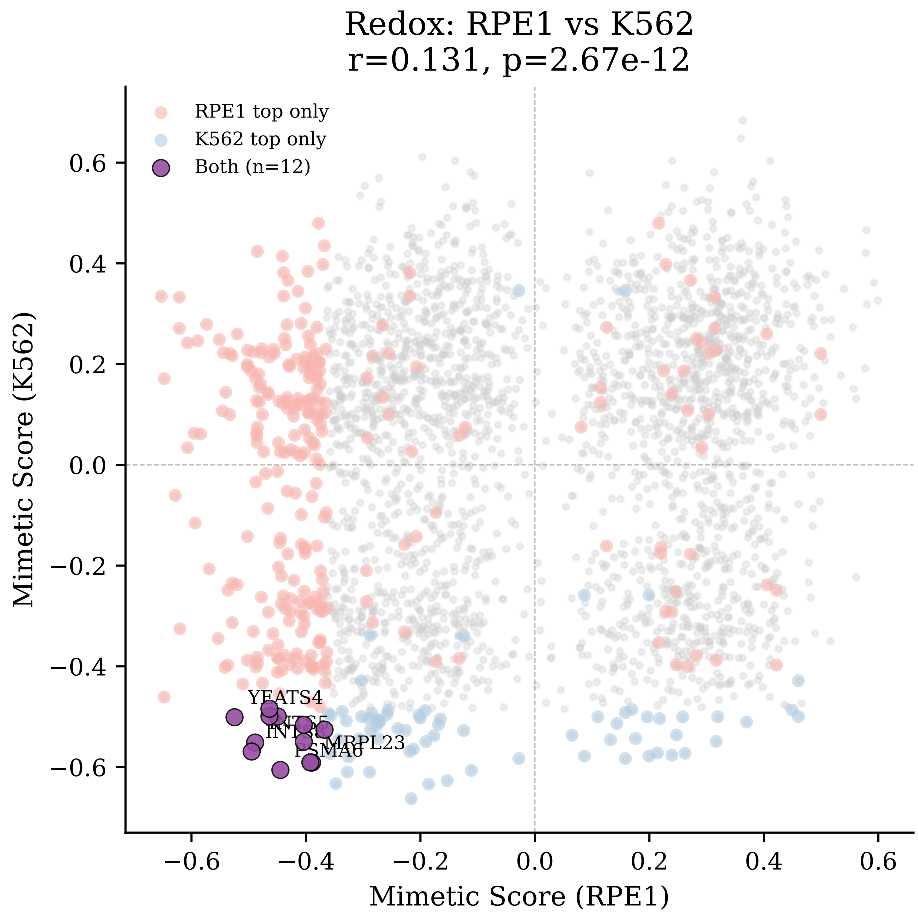


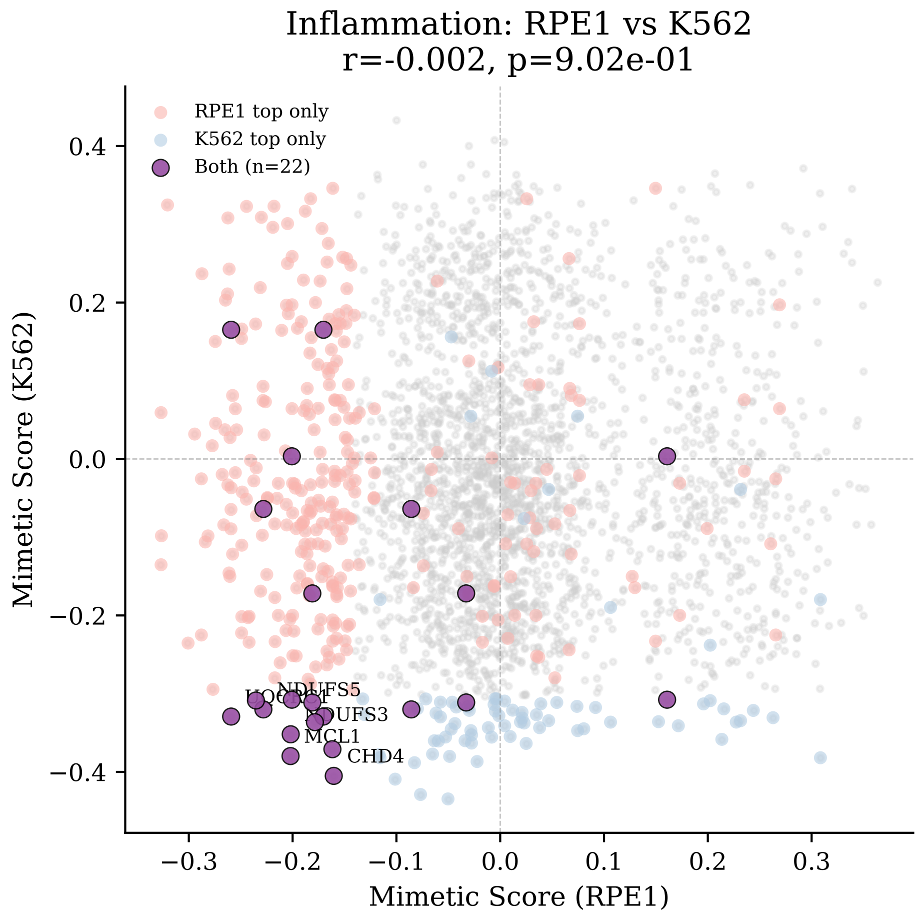
Figure 2: Concordance Heatmap



Concordance Scatter Plots **Age Concordance**



Redox Concordance



Inflammation Concordance

Convergent Genes

We identified **399 genes** that appear as top antagonists in multiple analyses (>=2 appearances). Top convergent genes include knockdowns that score highly across multiple pathways and/or both datasets.

Genes appearing in >=6 analyses: - See `convergent_genes_all.csv` for complete list

Output Files

Per-Analysis Outputs

Each analysis produces the following files in its respective directory:

```
results/perturbseq-analysis/
|-- Axes/
|   |-- Senescence/
|   |   |-- RPE1/backward/
|   |   |   |-- backward_Senescence_mimetics.csv
|   |   |   |-- top_200_mimetics.csv
|   |   |   |-- top_200_antagonists.csv
|   |   |   |-- Fig_Senescence_distribution.pdf
|   |   |   `-- Fig_Senescence_top_genes.pdf
```

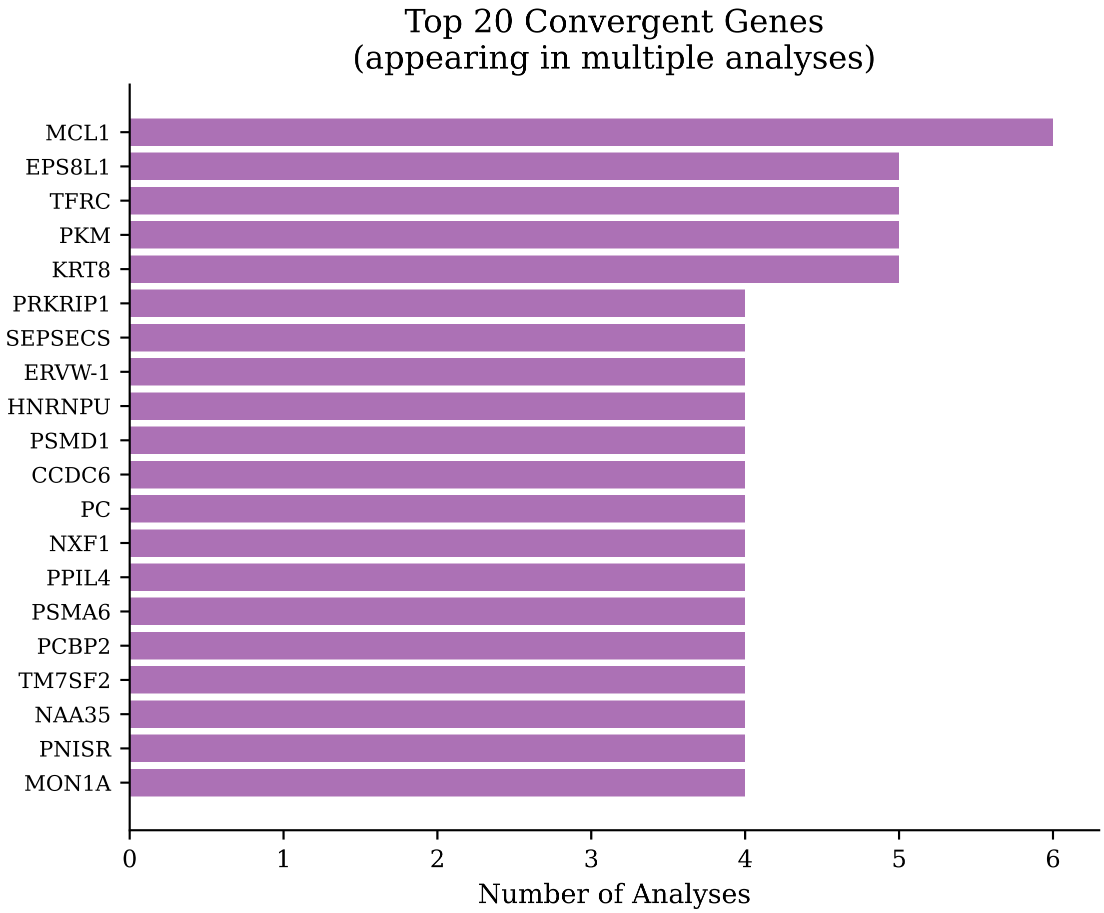


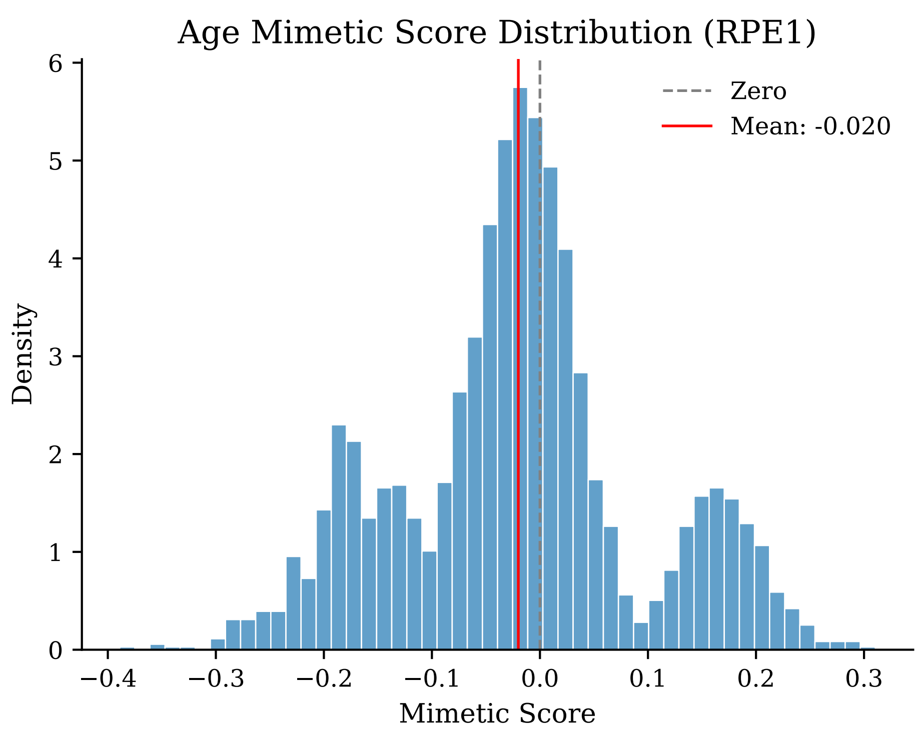
Figure 3: Convergent Genes Ranked

```
|   |   `-- K562_GWPS/backward/[same structure]
|   |-- Redox/[same structure]
|   `-- Inflammation/[same structure]
|-- AMD/
|   |-- RPE1/backward/[AMD-Macula, AMD-nonMacula results]
|   `-- K562_GWPS/backward/[same]
|-- concordance/
|   |-- rpe1_vs_k562_concordance.csv
|   |-- cross_geneset_overlaps_RPE1.csv
|   |-- cross_geneset_overlaps_K562_GWPS.csv
|   `-- convergent_genes_all.csv
`-- gene-sets/
    |-- registry.csv
    |-- coverage_in_perturbseq.csv
    `-- [individual gene set CSVs]
```

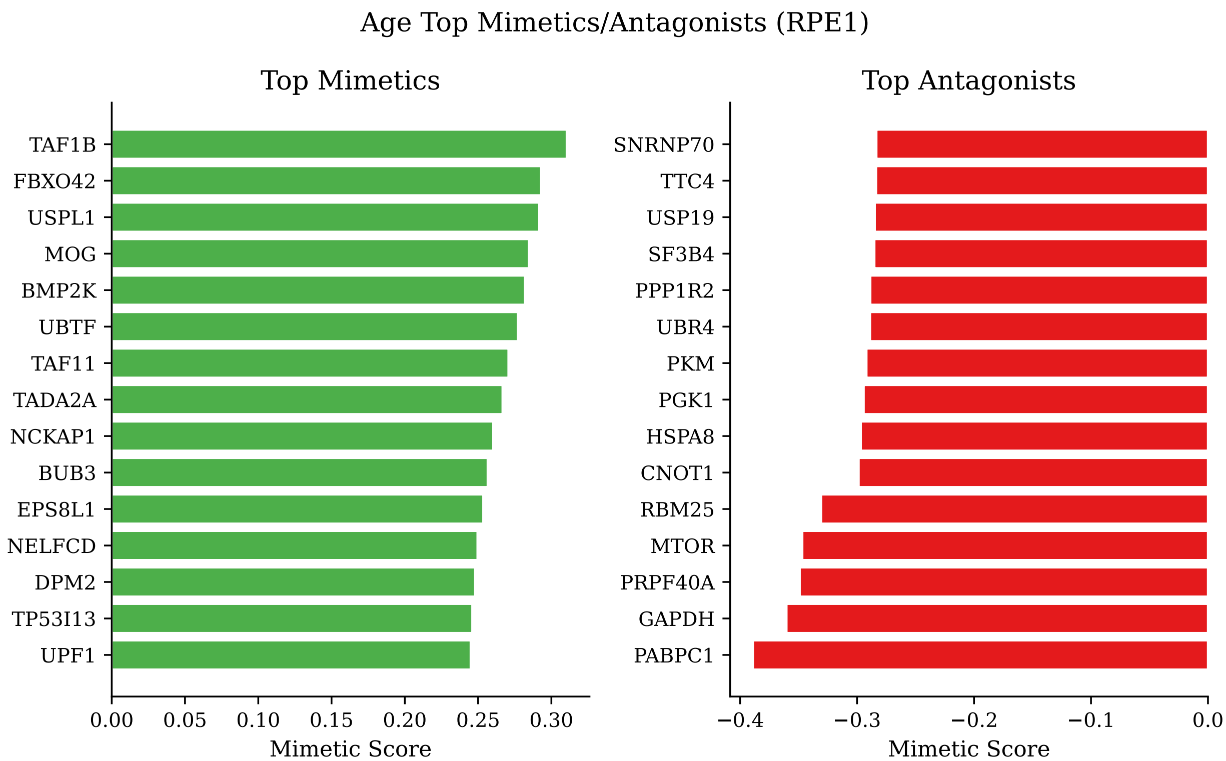
Interpretation

Age Signature

The Age signature analysis identifies knockdowns that reverse age-associated gene expression changes. The significant RPE1-K562 concordance (21 genes, p<1e-15) suggests conserved aging mechanisms across cell types.

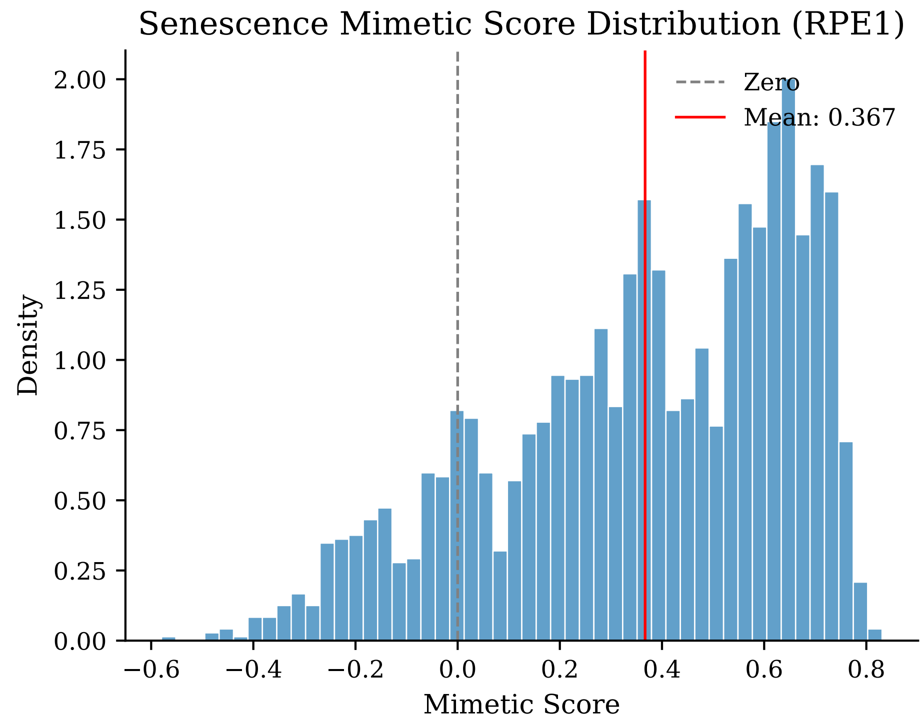


Age Analysis Results (RPE1)

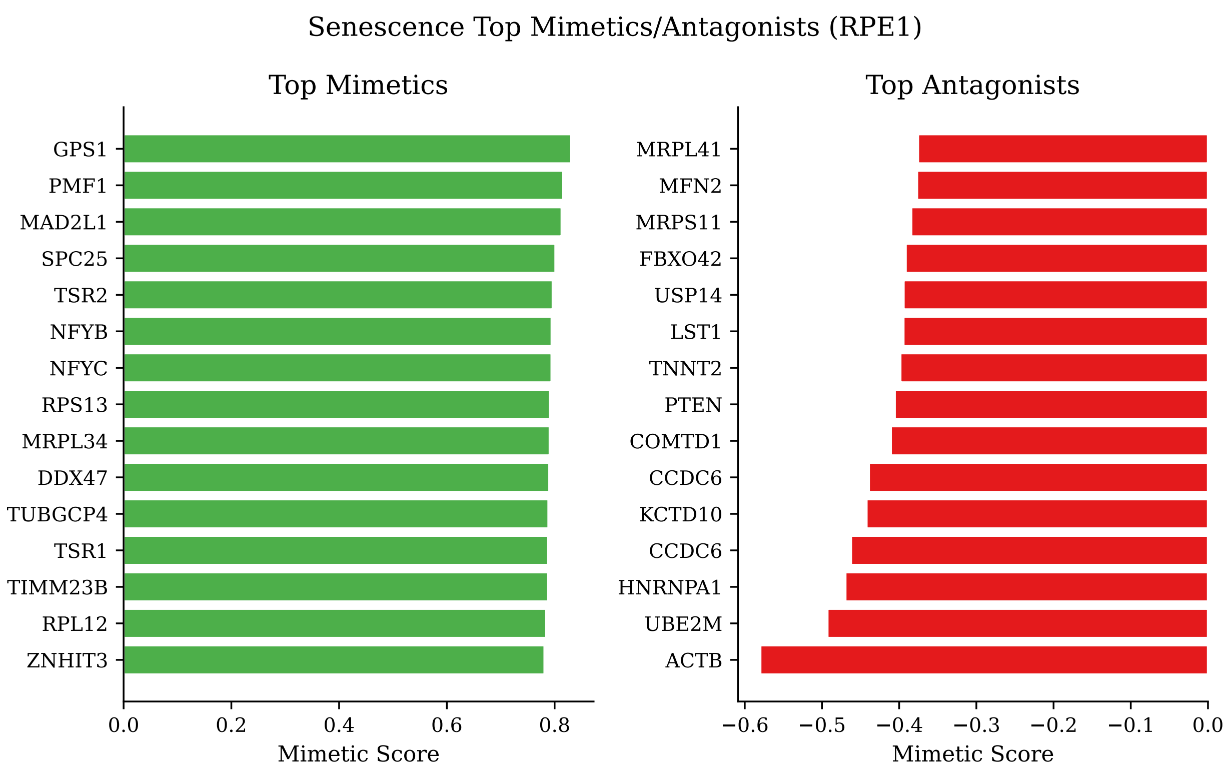


Senescence Axis

High coverage of Senescence-ANTI genes (92%) but lower Senescence-PRO coverage (32-43%) provides good detection of anti-senescence programs. The non-significant cross-dataset overlap may reflect cell-type-specific senescence programs.

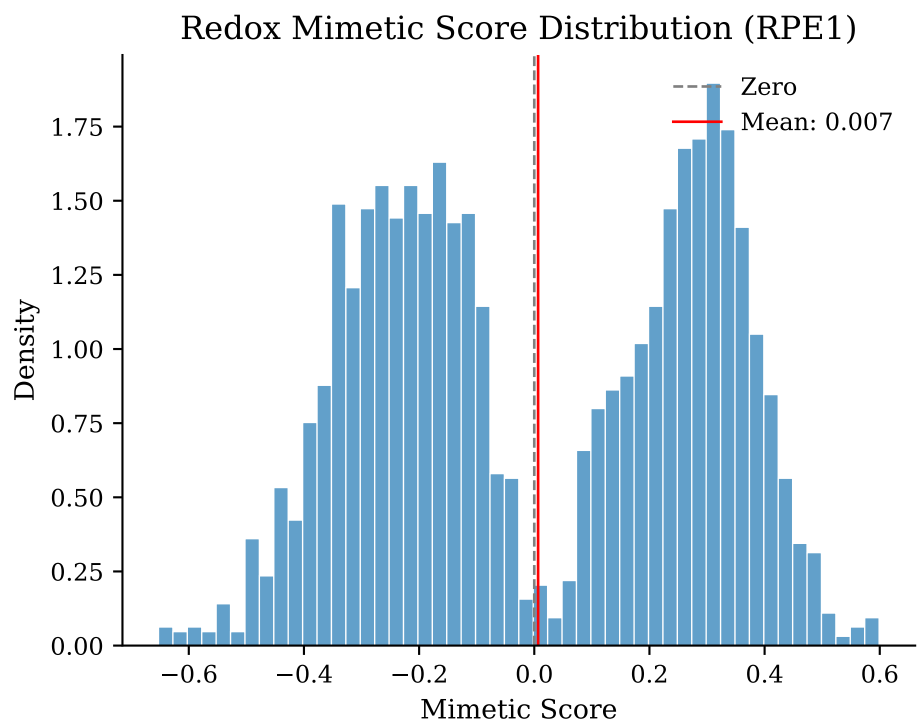


Senescence Analysis Results (RPE1)

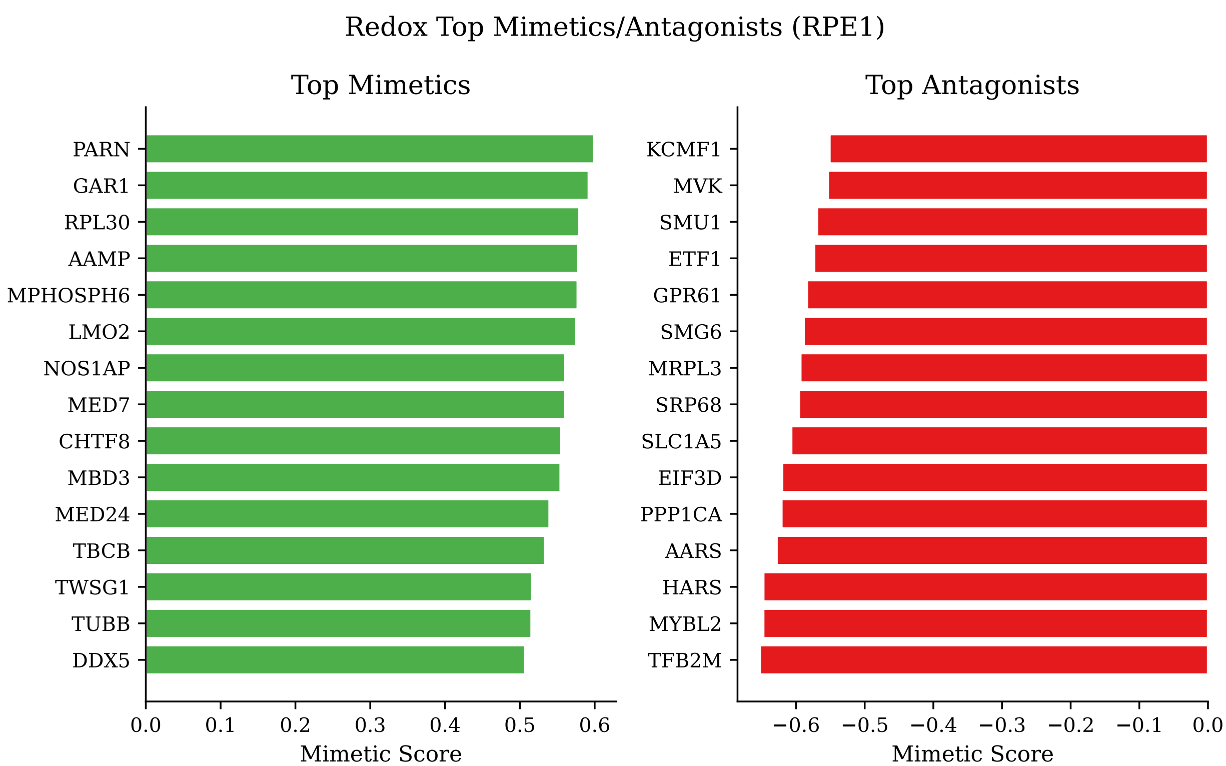


Redox Axis

Excellent coverage of Redox-PRO genes (94%) but limited Redox-ANTI coverage (56-63%) due to the small NRF2 target gene set. The significant cross-dataset concordance (11 genes, $p<5.7e\text{-}6$) indicates conserved oxidative stress response.

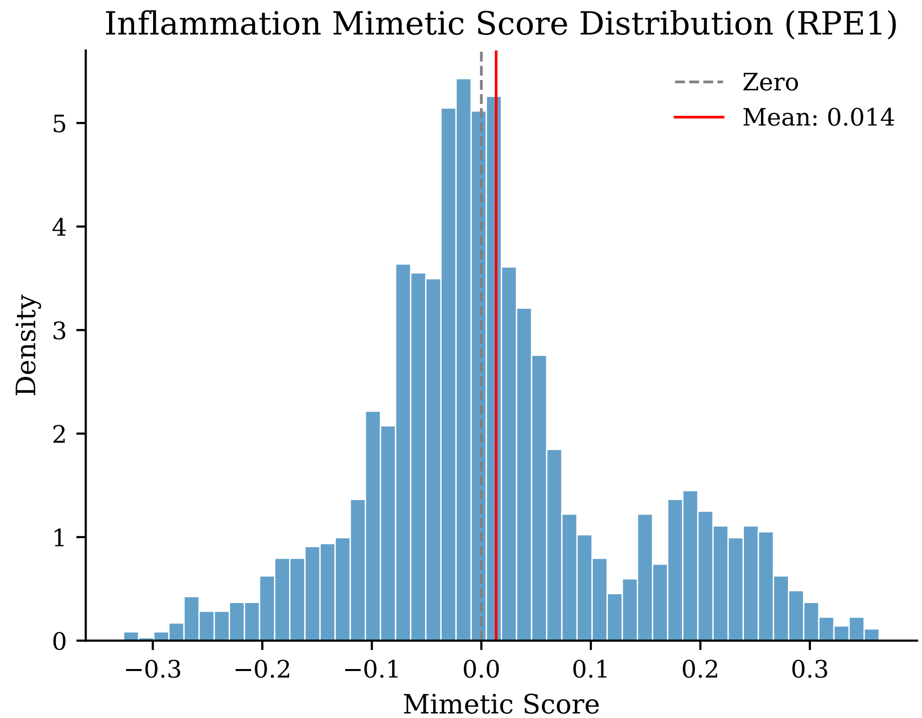


Redox Analysis Results (RPE1)

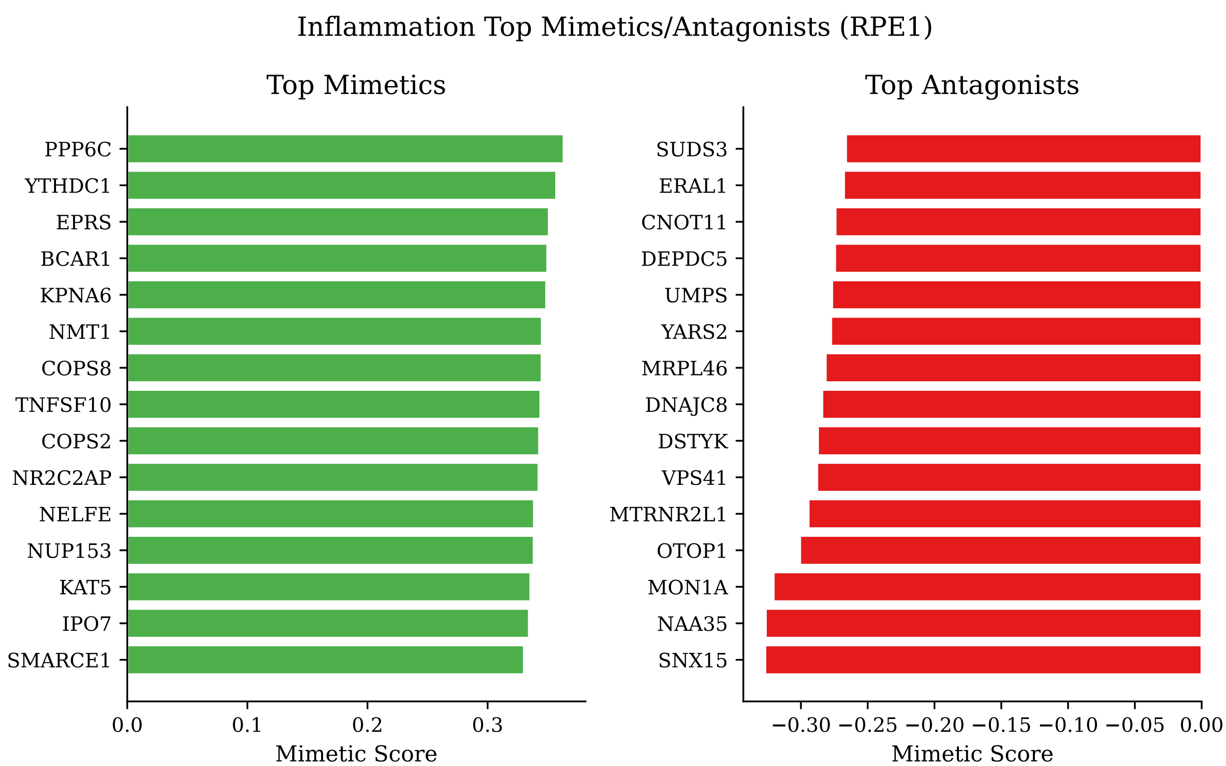


Inflammation Axis

Good coverage of Inflammation-PRO genes (36-47%) with significant cross-dataset overlap (9 genes, p<1.9e-4). The near-zero correlation (r=-0.001) suggests that while some antagonists overlap, the overall ranking differs between cell types.

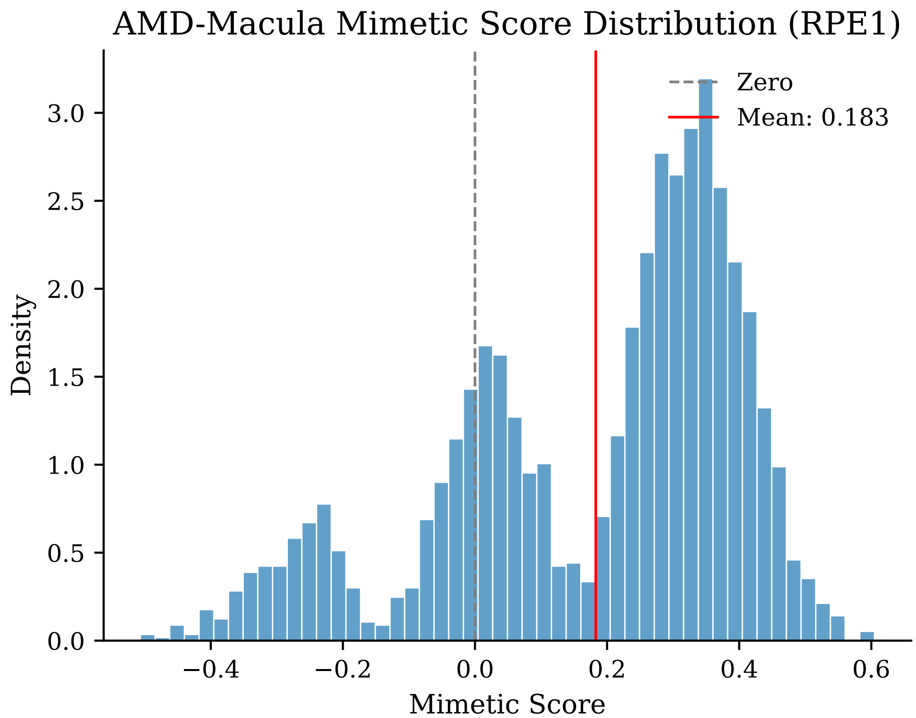


Inflammation Analysis Results (RPE1)

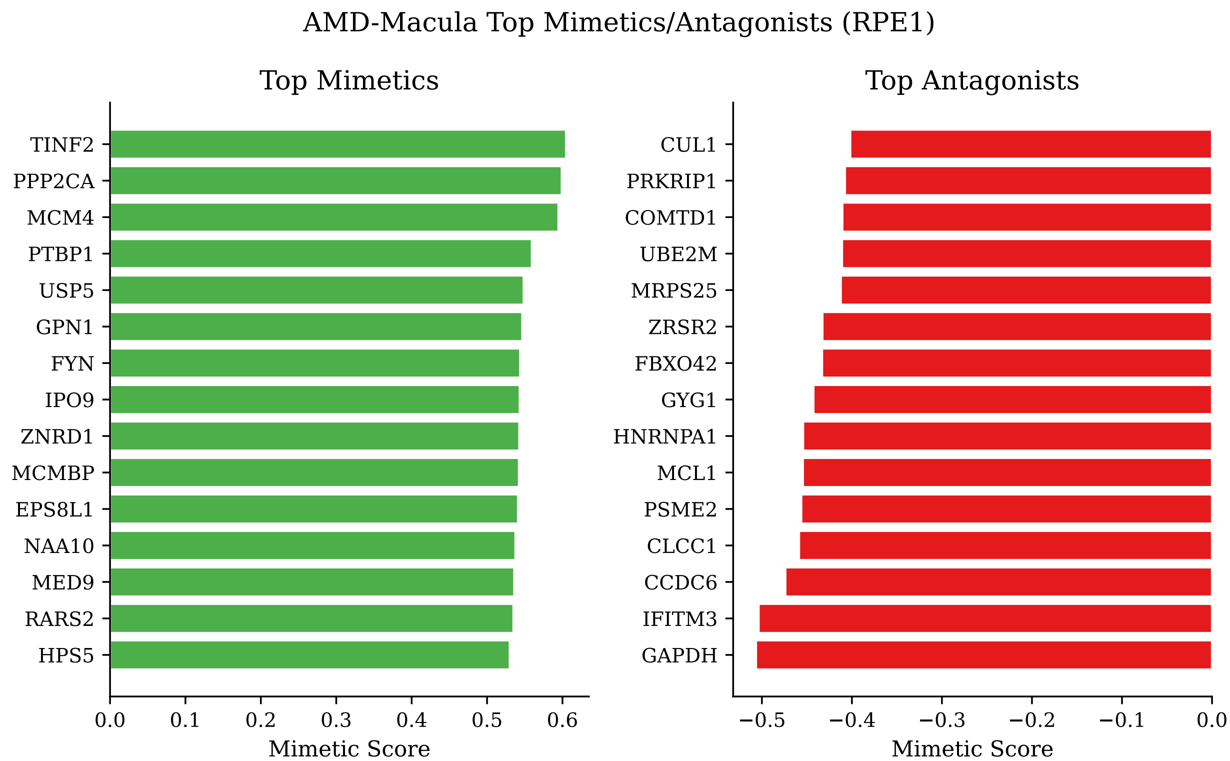


AMD Signatures

Low coverage for AMD gene sets reflects tissue-specific expression patterns in RPE that differ from cell line expression profiles. AMD-specific findings should be interpreted cautiously.



AMD-Macula Analysis Results (RPE1)



Conclusions

1. **Conserved pathways**: Age, Redox, and Inflammation pathways show significant cross-dataset concordance
2. **Cell-type specificity**: AMD and Senescence signatures show more cell-type-specific patterns
3. **Convergent targets**: 399 genes appear across multiple analyses as potential intervention targets
4. **Coverage limitations**: AMD-DOWN gene sets have low coverage, limiting sensitivity

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Technical Notes

- **Analysis runtime**: 7.5 minutes
- **Code location**: `code/perturbseq-analysis/`
- **Python environment**: `~/venvs/torch`
- **Key files**: `run_analysis.py` (orchestration), `enrichment_analysis.py` (GSEA scoring)